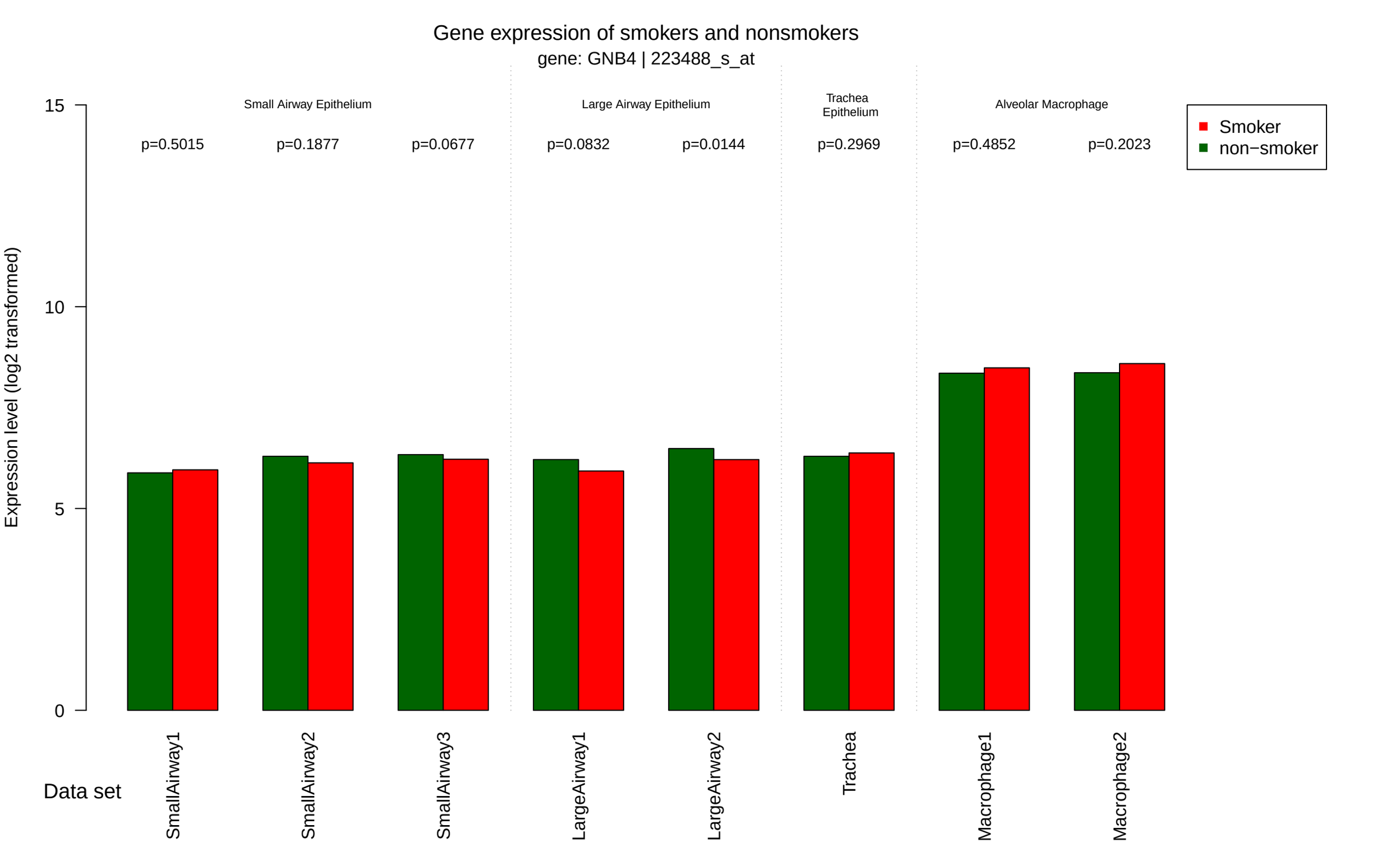
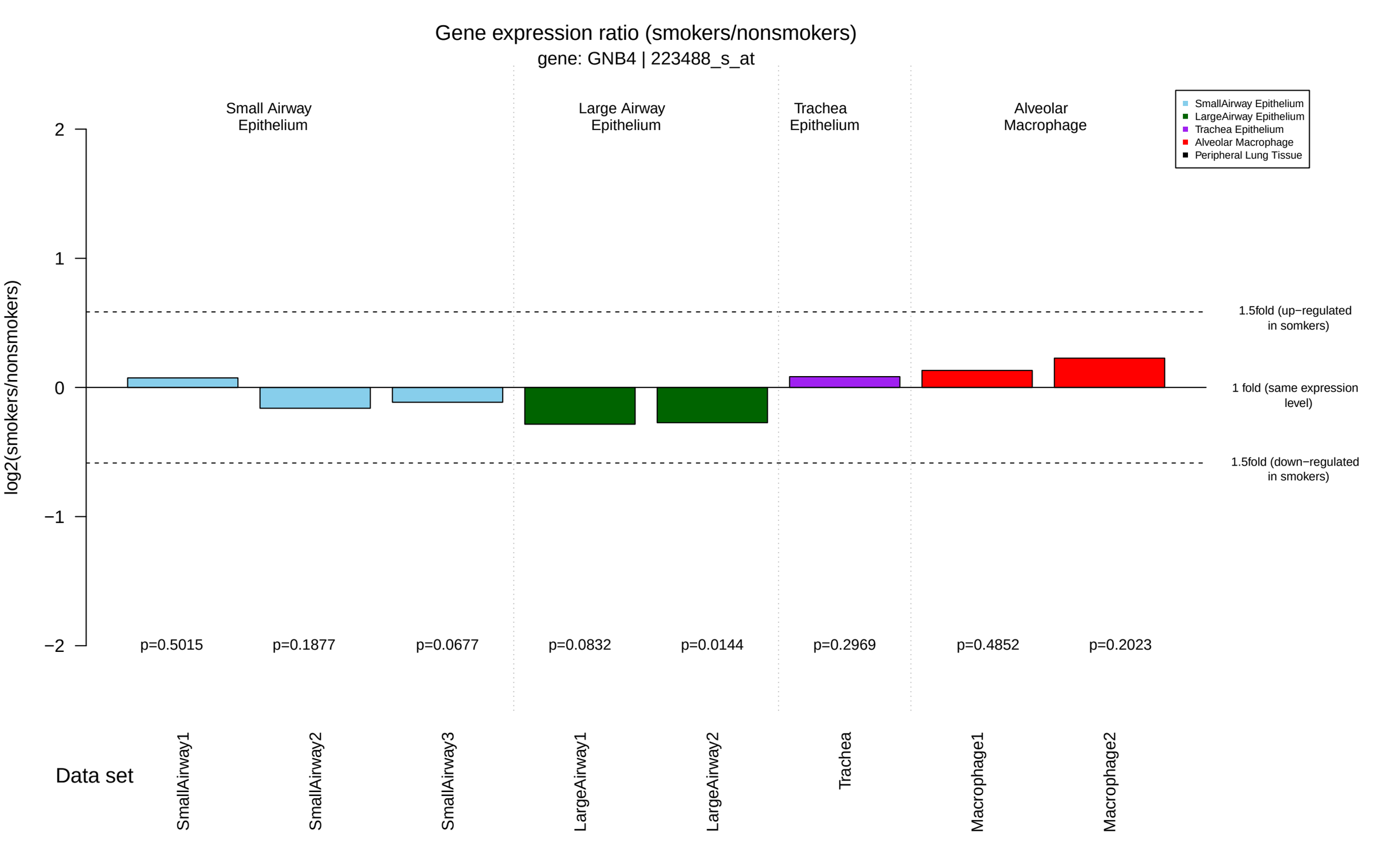
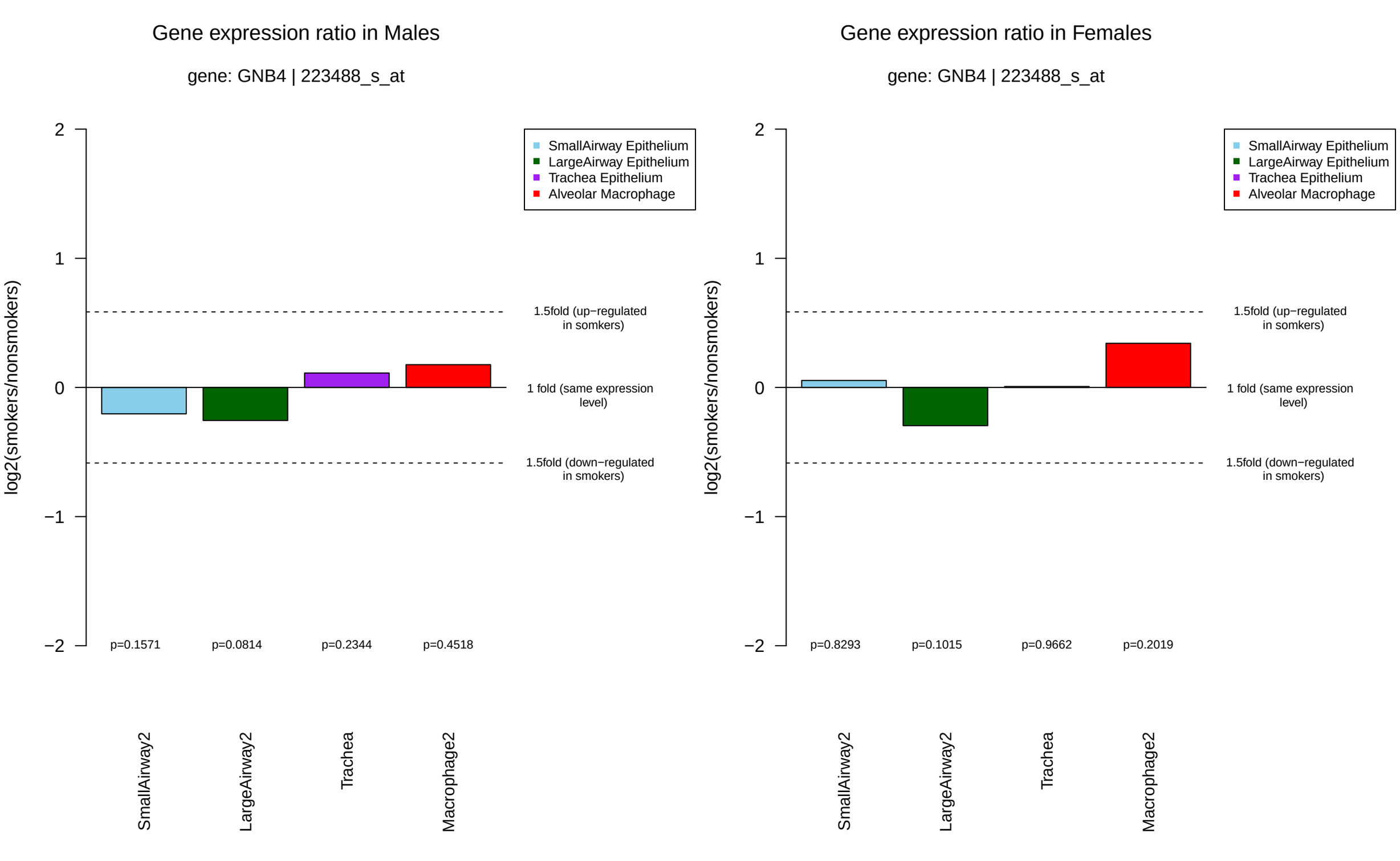
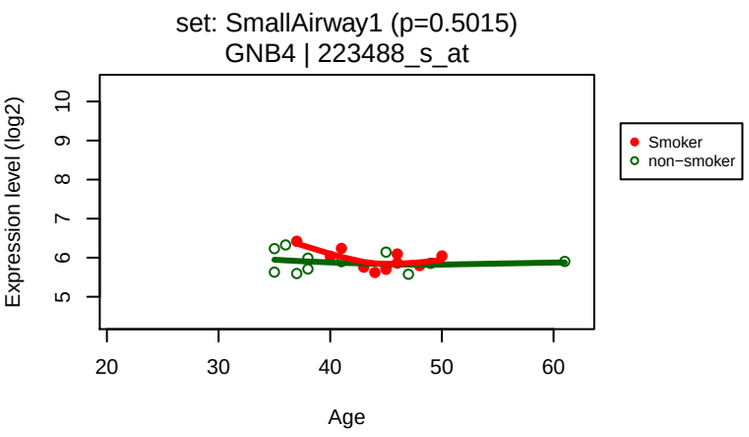
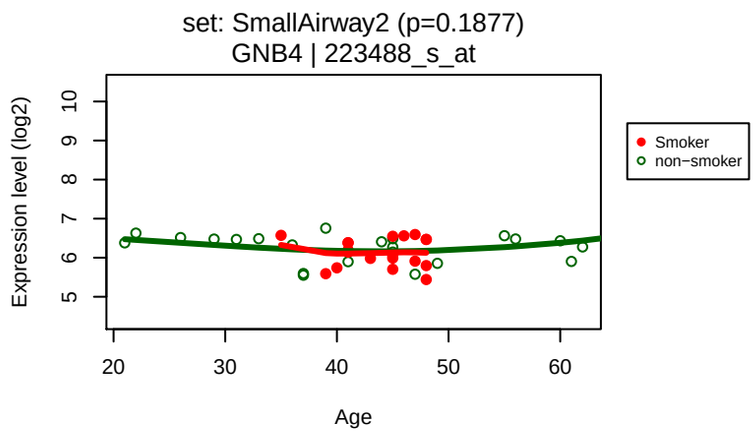
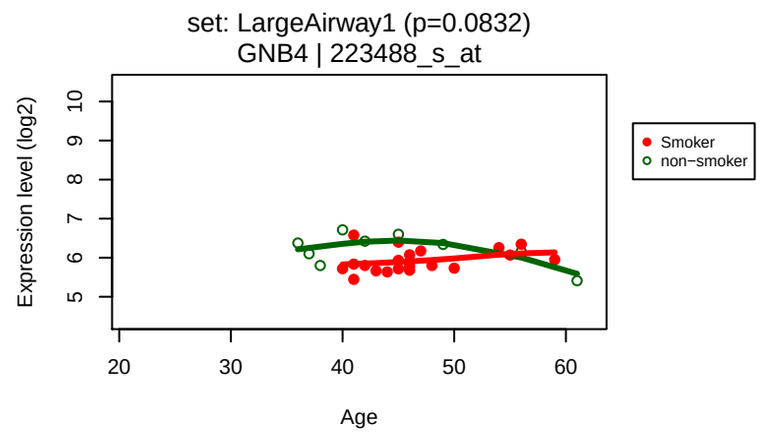

Gene expression of smokers and nonsmokers
gene: GNB4 | 223488_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5015
p=0.1877
p=0.0677
p=0.0832
p=0.0144
p=0.2969
p=0.4852
p=0.2023
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

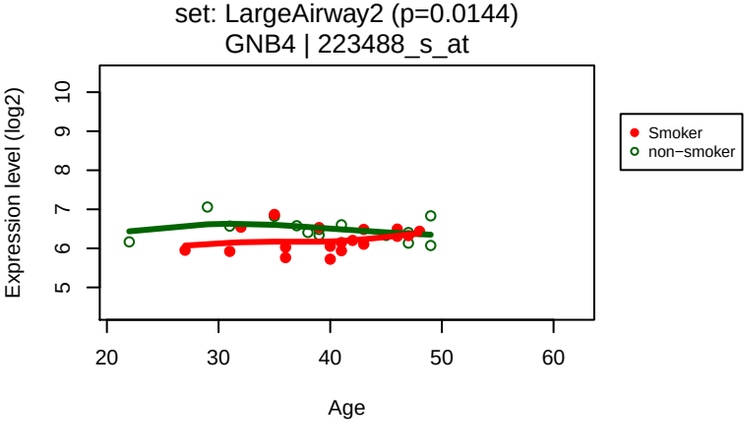

Gene expression ratio (smokers/nonsmokers)
gene: GNB4 | 223488_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5015
p=0.1877
p=0.0677
p=0.0832
p=0.0144
p=0.2969
p=0.4852
p=0.2023
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

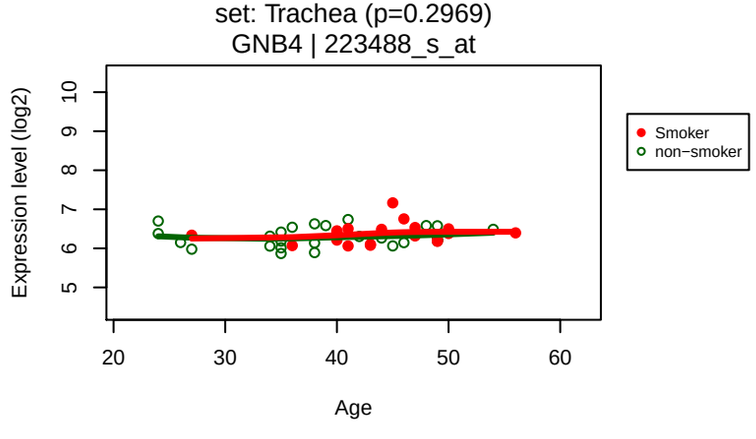

Gene expression ratio in Males
gene: GNB4 | 223488_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1571
p=0.0814
p=0.2344
p=0.4518
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNB4 | 223488_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8293
p=0.1015
p=0.9662
p=0.2019
SmallAirway2
LargeAirway2
Trachea
Macrophage2

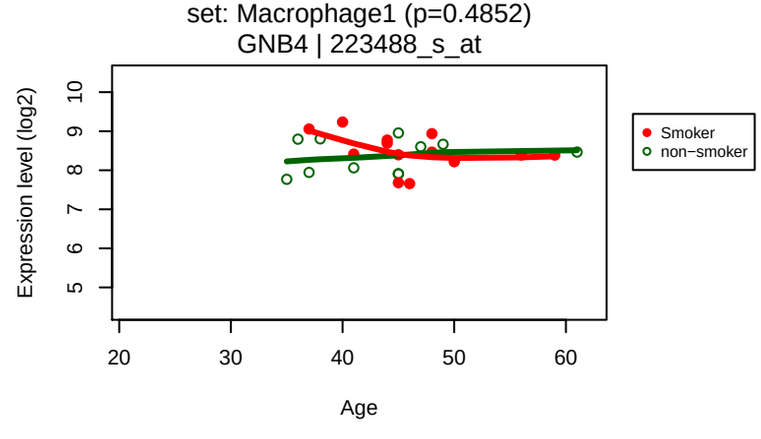

set: SmallAirway1 (p=0.5015)
GNB4 | 223488_s_at
Expression level (log2)
Age
Smoker
non−smoker

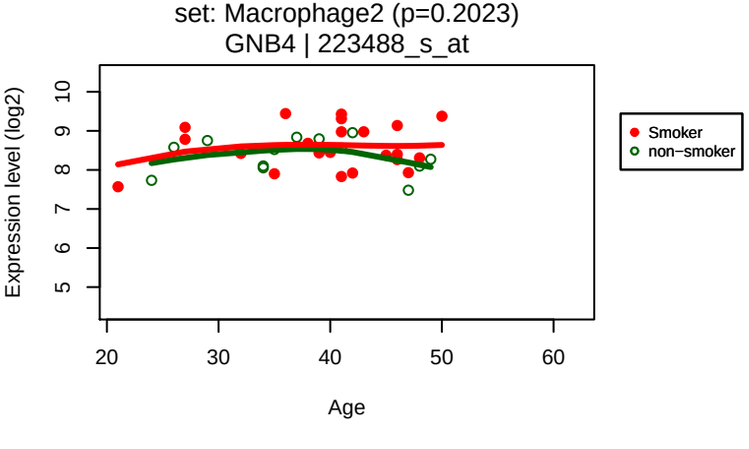

set: SmallAirway2 (p=0.1877)
GNB4 | 223488_s_at
Expression level (log2)
Age
Smoker
non−smoker

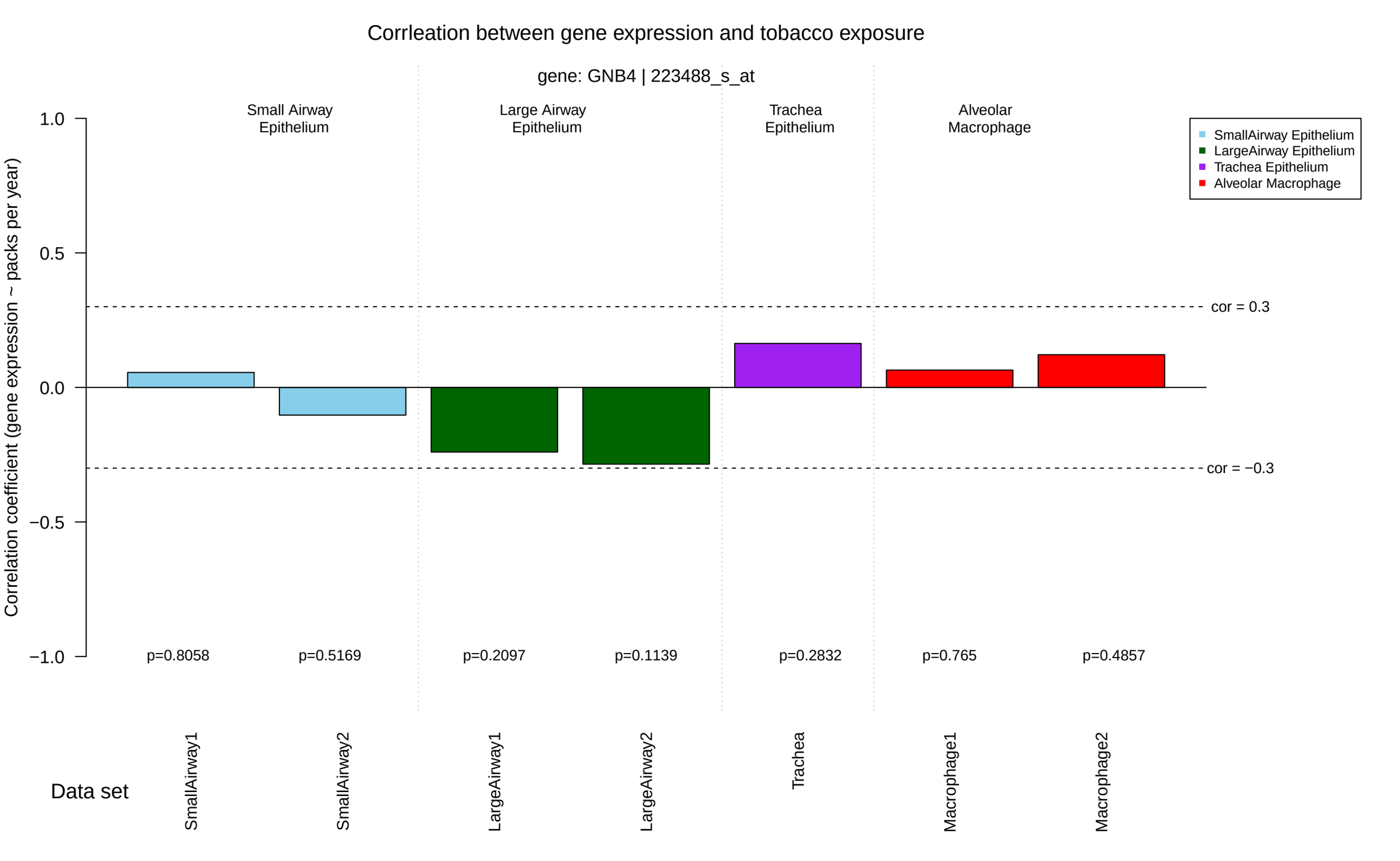

set: LargeAirway1 (p=0.0832)
GNB4 | 223488_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0144)
GNB4 | 223488_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2969)
GNB4 | 223488_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4852)
GNB4 | 223488_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2023)
GNB4 | 223488_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNB4 | 223488_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8058
p=0.5169
p=0.2097
p=0.1139
p=0.2832
p=0.765
p=0.4857
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2